

APPENDIX

Group : 9

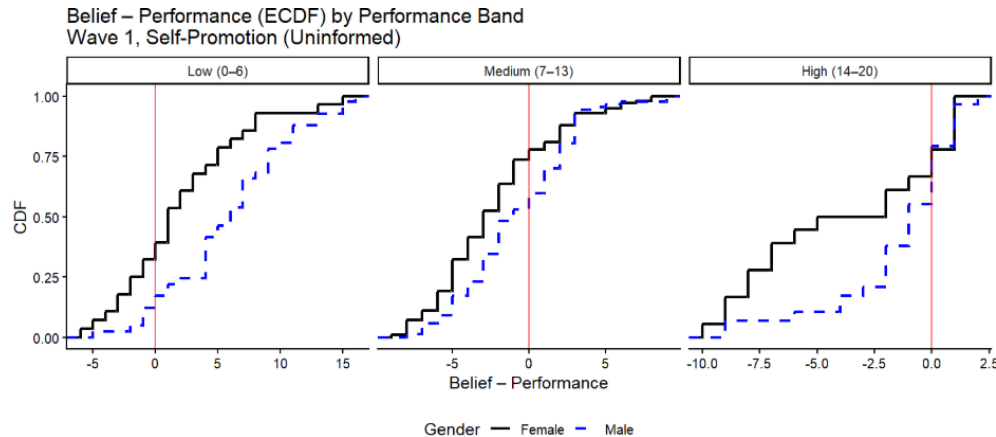
Team Members:

Aria Zhou, Jayachithra Thamizhmani, Srikirthi Srihari, Sri Sangeetha Sivashanmugan, Xiaohui Li

Section 1

Heterogeneity in the Gender Confidence Gap Across Performance Levels

perf_bin <fctr>	n_Female <int>	n_Male <int>	mean_gap_Female <dbl>	mean_gap_Male <dbl>	var_gap_Female <dbl>	var_gap_Male <dbl>	delta <dbl>	se_delta <dbl>
Low (0–6)	28	41	2.071429	5.951220	25.47619	24.897561	3.879791	1.2317149
Medium (7–13)	99	87	-2.242424	-0.816092	14.77737	12.779738	1.426332	0.5442057
High (14–20)	18	29	-3.888889	-1.482759	17.04575	7.472906	2.406130	1.0975758



Men consistently exhibit higher self-assessment relative to their objective performance across all ability levels.

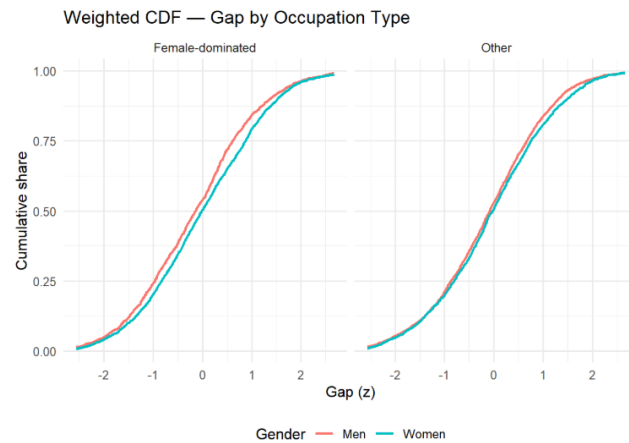
- Low-performance group (0–6): Both men and women tended to overestimate their performance; however, men’s overestimation was substantially larger on average (+5.95) compared to women (+2.07). This indicates that lower-performing men were markedly more confident than equally performing women.
- Medium-performance group (7–13): Both genders began to underestimate their performance, but women’s underestimation (–2.24) was nearly three times that of men (–0.82). The gender gap in self-assessment bias is smallest among participants with medium performance scores.
- High-performance group (14–20): The underestimation trend intensified, particularly among women (–3.89 vs. –1.48 for men). High-performing women were the most likely to underestimate their true ability.

Conclusion

We found that both men and women overestimate themselves in the low group, but women underestimate more strongly in the medium and high groups. The pattern suggests that the gender gap widens with ability or education level.

Section 2

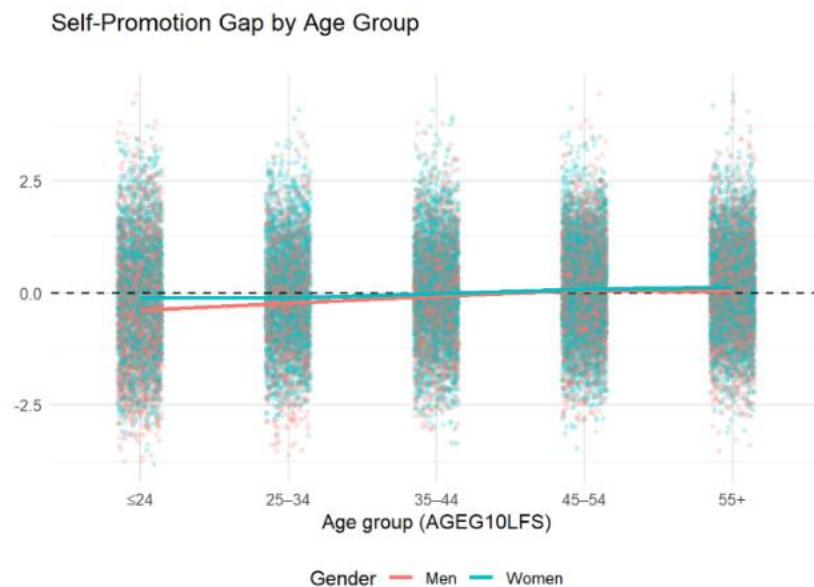
Extended Methodological Notes for Extension 1 (PIAAC Canada)



Self-Promotion Gap Distribution by Gender & Occupation - Weighted CDF

Interpretation

The distributions for female-dominated vs. other occupations overlap almost completely. However, women report higher self promotion in female dominated occupations.



Scatterplot of gap_z by Age Group and Gender

Interpretation

Gender differences are strongest among younger adults and flatten out for older groups.

Overall, gender differences in self-promotion are present, but they diminish with increasing education and increasing age, suggesting that experience and skill development reduce self-promotion disparities over time.

Model 1 — Baseline Gender Difference

Model 1 estimates the overall association between gender and self-promotion.

$$\text{gap}_z = \beta_0 + \beta_1(\text{Gender}) + \varepsilon$$

Numerical Results

- **Intercept:** -0.113
- **Gender coefficient:** +0.112
- **Interpretation**
 - Men (gender = 0) have an expected gap of **-0.11 SD**.
 - Women (gender = 1) have an expected gap of **≈ 0 SD**.
 - Women self-promote by **0.11 SD more** than men on average.

This difference is statistically significant but modest in practical magnitude. The same has been explained in the CDFs .

```
##
## == Model 1: gap_z ~ gender ==

summary(m1_gap)

##
## Call:
## svyglm(formula = gap_z ~ gender, design = des_brr)
##
## Survey design:
## svrepdesign.default(weights = as.formula(paste0("~", wt_final)),
##   repweights = rep_mat, type = "BRR", combined.weights = TRUE,
##   mse = TRUE, data = dat)
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.113443   0.002036  -55.71  <2e-16 ***
## gender       0.111709   0.002588   43.17  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for gaussian family taken to be 1.332219)
##
## Number of Fisher Scoring iterations: 2
```

TABLE 1: Regression Output for Model 1

Model 2 — Gender × Education Level

To test whether gender differences vary by education, we

$$\text{estimate: } \text{gap}_z = \beta_0 + \beta_1(\text{Gender}) + \beta_2(\text{Education Level}) + \beta_3(\text{Gender} \times \text{Education Level}) + \varepsilon$$

Key Coefficients (from your results):

- Gender: +0.201
- Gender × Medium: -0.020
- Gender × High: -0.153

Computed Gender Gaps by Education Level

- Low education: **0.201 SD**
- Medium education: **0.181 SD**
- High education: **0.048 SD**

Interpretation

Gender differences are **largest at low education** and shrink substantially with higher education. Among highly educated adults, the gender gap is **almost zero**.

```
##
## === Model 2: gap_z ~ gender * educ3 ===

summary(m_gap_educ)

##
## Call:
## svyglm(formula = gap_z ~ gender * educ3, design = des_brr)
##
## Survey design:
## svrepdesign.default(weights = as.formula(paste0("~", wt_final)),
##   repweights = rep_mat, type = "BRR", combined.weights = TRUE,
##   mse = TRUE, data = dat)
##
## Coefficients:
##
##               Estimate Std. Error
## (Intercept)    0.277965   0.005840
## gender
## educ3Medium education\n(High school / vocational) -0.432841   0.007080
## educ3High education\n(College / University)         -0.503300   0.006300
## gender:educ3Medium education\n(High school / vocational) -0.019922   0.009571
## gender:educ3High education\n(College / University)     -0.153063   0.009198
##
##               t value Pr(>|t|)
## (Intercept)    47.595 <2e-16 ***
## gender         24.040 <2e-16 ***
## educ3Medium education\n(High school / vocational) -61.132 <2e-16 ***
## educ3High education\n(College / University)       -79.885 <2e-16 ***
## gender:educ3Medium education\n(High school / vocational) -2.081  0.0409 *
## gender:educ3High education\n(College / University) -16.640 <2e-16 ***
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for gaussian family taken to be 1.291182)
##
## Number of Fisher Scoring iterations: 2
```

TABLE 2: Regression Output for Model 2

Model 3 — Gender × Age Group

To examine age-based heterogeneity:

$$\text{gap}_z = \beta_0 + \beta_1(\text{Gender}) + \beta_2(\text{Age Group}) + \beta_3(\text{Gender} \times \text{Age Group}) + \epsilon$$

Key Interaction Values (from your results):

- Gender (16–24): +0.2876
- Gender × 25–34: -0.1755
- Gender × 35–44: -0.2317
- Gender × 45–54: -0.2569
- Gender × 55+: -0.1925

Computed Gender Gaps by Age Group

- 16–24: **0.288 SD**
- 25–34: **0.112 SD**
- 35–44: **0.056 SD**
- 45–54: **0.031 SD**
- 55+: **0.095 SD**

Interpretation

The gender gap is **largest in early adulthood**, declines through midlife, and becomes **small or negligible** for most older groups.

```
##
## === Model 3: gap_z ~ gender * age_group ===

summary(m_gap_age)

##
## Call:
## svyglm(formula = gap_z ~ gender * age_group, design = des_brr)
##
## Survey design:
## svrepdesign.default(weights = as.formula(paste0("~", wt_final)),
##   repweights = rep_mat, type = "BRR", combined.weights = TRUE,
##   mse = TRUE, data = dat)
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -0.393866    0.004994  -78.86  <2e-16 ***
## gender          0.287060    0.007549   38.02  <2e-16 ***
## age_group25-34   0.168107    0.007137   23.56  <2e-16 ***
## age_group35-44   0.319049    0.006881   46.37  <2e-16 ***
## age_group45-54   0.443413    0.006117   72.49  <2e-16 ***
## age_group55+     0.418228    0.006168   67.81  <2e-16 ***
## gender:age_group25-34 -0.175336    0.010661  -16.45  <2e-16 ***
## gender:age_group35-44 -0.231720    0.010640  -21.78  <2e-16 ***
## gender:age_group45-54 -0.256953    0.008895  -28.89  <2e-16 ***
## gender:age_group55+  -0.192494    0.009066  -21.23  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for gaussian family taken to be 1.314456)
##
## Number of Fisher Scoring iterations: 2
```

TABLE 3: Regression Output for Model 3

Summary of Regression Findings

Across all models:

- Gender differences exist but remain **small in magnitude**.
- Education strongly moderates the gap — higher education → **smaller gender differences**.
- Age also moderates the gap — older groups → **smaller gender differences**.
- These effects are statistically significant but visually subtle, aligning with patterns seen in supporting plots.

Section 3

Extended Methodological Notes for Extension 2

How Education and Age Shape the Gender Confidence Gap

Results from the relationship between Gender gap and Education & Age of Original Paper:

- Both men and women show a declining belief–performance gap as education increases.
- Women consistently show a slightly larger belief–performance gap than men, but the difference narrows as education rises.
- Higher education is associated with a reduced gender gap.

This result is puzzling, in the U.S. high-school dataset we analyzed earlier, the gender gap is extremely small (about 0.1). This contradicts the decreasing gap pattern we observe here. Therefore, instead of treating education as a continuous scale, we next examine gender gaps across discrete education levels to see whether different education stages provide clearer clues.

Gender Gap Across Education Splits (High vs Low Groups)

To see how the gender gap changes when moving from a lower-education environment to a higher-education environment.

Model formula: Split-Sample Gender Gap Comparison Across Education Cutpoints

For each cutpoint $c \in \{1, \dots, 8\}$:

1. Split the sample $\rightarrow$ Low-Education group: $\text{Edu_ci} \leq c$, High-Education group: $\text{Edu_ci} > c$
2. In two simple regressions:
Low group : $Y_i = \beta_{0L} + \beta_{1L} \text{Female}_i + \epsilon_i$ & High group : $Y_i = \beta_{0H} + \beta_{1H} \text{Female}_i + \epsilon_i$
3. Compare gender gaps:
 $\Delta \text{Gap} = B1H - B1L \rightarrow \text{Gap (high-education grp)} - \text{Gap (Low-education grp)}$

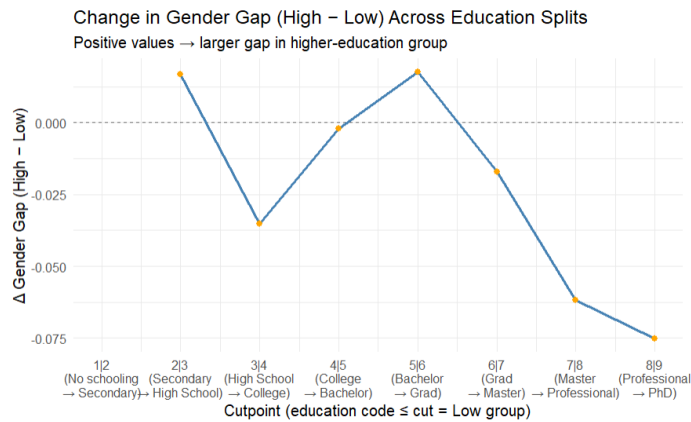


Figure: Change in Gender Gap (High – Low) Across Education Splits

Result

Across eight education splits, the pattern is not monotonic. A clear turning point appears at the High School cutoff (3 $\rightarrow$ 4).

After this point, the gap rises when entering college, then gradually declines again as education increases.

Conclusion

The gender gap is smallest at the High School group, matching our U.S. high-school dataset (≈ 0.1). It widens when entering college, possibly because women begin to feel stronger social pressure on self-evaluation. As education increases further, the gap narrows again, suggesting improved confidence and self-evaluation.

Section 4

Extension 3 - Early Educational Evidence of the Gender Confidence

This extension explores whether the gender confidence gap observed in adult labor-market behavior is already evident during earlier educational stages. Using the U.S. HSLS:09 dataset, we examine links between performance, self-confidence, and gender among high school students. All analyses were run in NCES PowerStats, which provides weighted results without releasing microdata.

4.1 Data Overview: Sources, Sample construction & Variables

Data Source

- **Data Used:** HSLS:09 (NCES), link: <https://nces.ed.gov/datalab/powerstats/72-high-school-longitudinal-study-of-2009>
- **Access:** Via NCES PowerStats, which supports weighted analyses while protecting respondent confidentiality.
- **Format:** Internally processed via PowerStats platform.
- **Focus:** 10th-grade student sample ($N \approx 21,000$) measuring academic performance (GPA, test scores), math self-efficacy and gender designed to capture early cognitive and motivational differences across educational backgrounds.

Sample Construction

Data Cleaning: Variables cleaned and standardized using NCES documentation within the PowerStats environment.

Interpretation Constraints

- Access: PowerStats restricts access to microdata and limits custom diagnostics.
- Measurement: Math self-efficacy is self-reported and may not perfectly reflect true confidence.

Despite these limits, HSLS:09 provides credible early evidence that gender confidence gaps emerge before adulthood, supporting the robustness of our replicated findings.

Specification

PowerStats uses weighted least squares (WLS), conceptually equivalent to the OLS with performance fixed effects used in Exley & Kessler (2022):

$$Y_i = \beta_0 + \beta_1 \times 1(\text{gender}) + \beta_2 \times 2(\text{performance}) + \varepsilon$$

- Female : Indicator variable (1 = Female, 0 = Male)
- Performance : GPA
- Y = Math self-efficacy

Computation occurs on NCES servers for privacy, but both approaches estimate the gender effect while controlling for ability.

Variable Definition

Study Variable	Dataset Variable	Definition	Type
performance	perf_score	Evaluated performance score (0–100)	Continuous
GPA (Performance Score)	X3T_GPA	Average GPA for all academic 10th-grade courses (0–4 scale)	Continuous
Math Self-Efficacy	X1_MATHSEFF	Scale of confidence in one’s mathematics ability	Continuous
Gender	X1_SEX	Dummy: 1 = female, 0 = male	Binary

4.2 Analysis and Results

We use descriptive correlations and weighted comparisons to examine how gender, performance, and self-confidence relate in HSLS:09, including:

- Weighted correlations between GPA, math self-efficacy, and gender.
- Comparisons of these patterns with the original Exley & Kessler (2022) findings.

Correlation matrix for X3T GPA for all academic 10th grade courses, X1 Student's sex and X1 Scale of student's mathematics self-efficacy.

		X3T GPA for all academic 10th grade courses	X1 Student's sex		X1 Scale of student's mathematics self-efficacy
			Male	Female	
X3T GPA for all academic 10th grade courses		1.000			
X1 Student's sex	Male	-0.196	1.000		
	Female	0.196	-1.000	1.000	
X1 Scale of student's mathematics self-efficacy		0.216	0.100	-0.100	1.000

Figure 3: Correlation Between Academic Performance, Gender, and Math Self-Efficacy (HSLs:09, NCES PowerStats).

Comparison and Findings

Dependent Variable	Data Source	β / r Estimate	Interpretation	Related to Original Findings
Self-Evaluation/ Performance Score	Exley & Kessler (2022) – Experimental Data	$\beta = -12.68^{***}$ (2.96)	Female participants rated their own performance $\approx$ 12.7 points lower than males with equal ability.	Baseline experimental evidence of confidence gap.
Math Self-Efficacy (Confidence Score)	HSLs:09 – Educational Data	$\Delta \approx -0.10$ (correlation $r \approx -0.10$)	Female students show slightly lower confidence despite equal or higher GPA; $r \approx -0.10$ indicates a weak negative relationship, meaning the gender confidence gap exists but is small.	Same direction, smaller magnitude — weak correlation confirms the gap but much less pronounced in the educational context.
Interpretation - Both datasets show a gender confidence gap. In the experiment, women underrated their performance by about 12.7 points, indicating a strong self-evaluation bias. In HSLs:09, the gap is much smaller ($r \approx -0.10$), suggesting that confidence differences exist but are far weaker in high school than in the experimental setting.				

Section 5

Reproducibility

5.1 Data-Usage Note

All datasets used in this project were accessed through publicly available, non-restricted sources and used solely for academic, non-commercial purposes.

- Replication & Extension 2 (Original Experiment Data):
Official replication files from *The Gender Gap in Self-Promotion* (Exley & Kessler, 2022), obtained through the Harvard Dataverse (open access, provided as .dta files).

- Extension 1 (PIAAC Canada Data):
OECD Programme for the International Assessment of Adult Competencies (PIAAC), Cycle 1 – Canada (2012). Publicly accessible through the OECD Data Explorer and national microdata portals, subject to standard public-use licensing.
- Extension 3 (HSL:09 High School Data):
U.S. NCES High School Longitudinal Study of 2009 (HSL:09), accessed via NCES PowerStats. All analysis followed NCES confidentiality protections; individual-level microdata were not downloaded or viewed.

5.2 Data dictionary

Dataset metadata

Dataset	File path	Source	Format	License / Access Notes
<i>Replication — Original Experiment</i>	<i>data/main.dta</i>	<i>Harvard Dataverse (Exley & Kessler, 2022)</i>	<i>.dta</i>	<i>Open access (Harvard Dataverse)</i>
<i>Extension 1 — PIAAC Canada (Cycle 1)</i>	<i>data/prgcanada.csv</i>	<i>OECD Data Explorer / Statistics Canada microdata</i>	<i>.csv</i>	<i>Public microdata (requires user agreement)</i>
<i>Extension 3 — HSL:09 (NCES)</i>	<i>Accessed via NCES PowerStats (no raw file downloaded)</i>	<i>NCES DataLab: HSL:09 PowerStats</i>	<i>Web tool (no microdata)</i>	<i>Restricted microdata; accessible via PowerStats only</i>

Variable Usage

Replication and Extension 2 – Original paper dataset

Variable	Description	Type	Value / Notes	Source	Used in Chunk / Step
female	Gender indicator	Binary	0 = male, 1 = female	Dataset	1.2.1- OLS regression, summary tables
perf	Actual performance score	Numeric	0–20	Dataset	1.2.1 -OLS regression, creating perf_bucket, summary
belief	Self-evaluated performance	Numeric	0–20	Dataset	1.2.1 -OLS regression, gap calculation
edlevel3	Education category	Categorical	1 = Low, 2 = Medium, 3 = High	Dataset	1.2.1 - Covariate in regressions

numeracy	PIAAC numeracy score	Continuous	Plausible values	Dataset	1.2.1 - Covariate in regressions
numeracy_use	Self-reported numeracy use	Continuous	1–5 Likert scale	Dataset	1.2.1 - Covariate in regressions
gap	Self-assessment bias (belief – perf)	Numeric	Can be centered or z-scored	Derived	1.2.2 -OLS regression, summary tables
perf_bucket	Performance bucket (from perf)	Categorical	E.g., 0–5, 6–10, 11–15, 16–20	Derived	1.2.3 - Regression by bucket
apply	Application indicator	Binary	0 = did not apply, 1 = applied	Derived	1.3.1 -OLS regression for application outcome
success	Success indicator	Binary	0 = not,successful, 1 = successful	Derived	1.3.1 -OLS regression for success outcome
pred_perf	Predicted performance from regression	Numeric	Continuous numeric score	Derived	1.4.1 - Computing residuals, diagnostics
pred_gap	Predicted gap from regression	Numeric	Continuous numeric score	Derived	1.4.1 - Computing residuals, diagnostics
resid_perf	Residual of performance (perf – pred_perf)	Numeric	Continuous numeric score	Derived	1.4.2- Diagnostics, summary tables

Extension 1 – PIAAC data

Variable	Description	Type	Value / Notes	Source	Used in Chunk / Step
female	Gender indicator	Binary	0 = male, 1 = female	Dataset	1.2.1 – OLS regression, summary tables
perf	Actual performance score	Numeric	0–20	Dataset	1.2.1 – OLS regression, creating perf_bucket summary

belief	Self-evaluated performance	Numeric	0–20	Dataset	1.2.1 – OLS regression, gap calculation
edlevel3	Education category	Categorical	1 = Low, 2 = Medium, 3 = High	Dataset	1.2.1 – Covariate in regressions
numeracy	PIAAC numeracy score	Continuous	Plausible values	Dataset	1.2.1 – Covariate in regressions
numeracy_use	Self-reported numeracy use	Continuous	1–5 Likert scale	Dataset	1.2.1 – Covariate in regressions
gap	Self-assessment bias (belief – perf)	Numeric	Can be centered or z-scored	Derived	1.2.2 – OLS regression, summary tables
perf_bucket	Performance bucket (from perf)	Categorical	0–5, 6–10, 11–15, 16–20	Derived	1.2.3 – Regression by bucket
apply	Application indicator	Binary	0 = did not apply, 1 = applied	Derived	1.3.1 – OLS regression for application outcome
success	Success indicator	Binary	0 = not successful, 1 = successful	Derived	1.3.1 – OLS regression for success outcome
pred_perf	Predicted performance from regression	Numeric	Continuous numeric score	Derived	1.4.1 – Computing residuals, diagnostics
pred_gap	Predicted gap from regression	Numeric	Continuous numeric score	Derived	1.4.1 – Computing residuals, diagnostics

Section 6

Division of Labour Statement

Name	Contribution
Aria Zhou	Worked on replication of the original study (<i>The Gender Gap in Self-Promotion</i>), including all diagnostic checks to ensure data and code integrity and heterogeneity analysis of the replication.

Jayachithra Thamizhmani	Drafting and compiling the Project Report and the reproducibility Appendix. Also worked on Extension 2 - education, age shape the gender confidence gap analysis.
Srikirithi Srihari	Worked on Extension 1 using the PIAAC Canada dataset, and developed a single executable R script consolidating all analyses for reproducibility.
Sri Sangeetha Sivashanmugan	Also worked on Extension 1 analysis and curated the final presentation slides for communicating the project findings.
Xiaohui Li	Worked on replication of the original study and Extension 2 - the analysis of how education, age influences the gender confidence gap. Extension 3 - Early Educational Evidence of the Gender Confidence.

Section 7

AI Use Declaration

Our team used AI tools (ChatGPT by OpenAI) only for non-analytical support, including:

- improving clarity and organization of written sections,
- helping with grammar, editing, and formatting of tables/figures,
- generating boilerplate explanations (e.g., definitions, standard methodological descriptions),
- assisting in restructuring the narrative for readability.

All code, statistical analyses, replication procedures, diagnostics, data processing, and extension analyses were written, executed, and verified by the team.

AI was *not* used to generate, manipulate, or fabricate any empirical results, nor to run statistical models or produce code outputs.

All numerical results come directly from our R scripts and the official datasets.